

Def Let V, W be any vector spaces. A map $T: V \rightarrow W$ is called linear if

$$\left. \begin{array}{l} \textcircled{1} T(v_1 + v_2) = T(v_1) + T(v_2) \\ \textcircled{2} T(cv_1) = cT(v_1) \end{array} \right\} \text{ for all } v_1, v_2 \in V, c \in \mathbb{R}$$

It is easy to see that $\left\{ \begin{array}{l} \ker T = \{v \mid T(v) = 0\} \text{ is a subsp of } V \\ \operatorname{im} T = \{Tw \mid v \in V\} \text{ is a subsp. of } W \end{array} \right.$

Thm For a linear map $T: V \rightarrow W$ where V is fdim V -space,
 $\dim \ker T + \dim \operatorname{im} T = \dim V$

"
nullity of T

"
rank of T

[Pf: below]

Good to remember for any linear map $V \xrightarrow{T} W$

(i) Basic example: For an $r \times c$ matrix A , $T(x) = Ax$ is a linear map from $\mathbb{R}^c$ to $\mathbb{R}^r$.

We will see Any linear $T: \underbrace{V \rightarrow W}_{\text{fdim vsp}}$, after choosing a basis for V & one for W

"looks like $x \mapsto Ax$ " (precise meaning later)

Still abstract viewpoint is very useful

(recall e.g. derivative map)

(ii) Any linear T preserves linear combinations i.e.

$$T(\sum c_i v_i) = \sum c_i T(v_i) \quad \text{Easy consequences:}$$

For a given set $S \subset V$,

$S \text{ spans } V \Rightarrow T(S) \text{ spans } T(V)$. In fact $T(\operatorname{Span} S) = \operatorname{Span}(T(S))$

$S \text{ lin indep} \Leftarrow T(S) \text{ lin indep}$

Proof of thm We will show

$$\underbrace{(\text{basis of } \ker T)}_{\cap} \cup \underbrace{(\text{a disjoint set of size } \dim \operatorname{Im} T \text{ in } V)}_{\cap} = \underbrace{(\text{a basis of } V)}_{\cap}$$

1) Take a basis $k_1, \dots, k_s$ of $\ker T$

2) Extend to get a basis $k_1, k_2, \dots, k_s, v_1, \dots, v_r$ of V

As $\dim \ker T = s$, $\dim V = s+r$, we are reduced to showing $\dim \operatorname{Im} T = r$

3) Obvious guess/claim: (which will prove the thm) :

$T(v_1), \dots, T(v_r)$ is a basis of $\operatorname{Im}(T)$

• These vectors span $\operatorname{Im} T$ pf: We know that, since

$\{k_1, \dots, k_s, v_1, \dots, v_r\}$ spans V , $T(k_i) \neq T(v_j)$ span $T(V)$

i.e. anything in the image = $T(\text{some } v)$

as each $T(k_i) = 0$

$$= T\left(\sum_{i=1}^s a_i k_i + \sum_{j=1}^r b_j v_j\right) \stackrel{\text{by linearity of } T}{=} \sum a_i T(k_i) + \sum b_j T(v_j) \in \operatorname{span} T(v_j)\text{'s.}$$

• $T(v_1), \dots, T(v_r)$ are lin indep- pf. Suppose $\sum_{i=1}^r c_i T(v_i) = 0$

$$\therefore \text{By linearity of } T, \text{ LHS} = T\left(\sum_i c_i v_i\right) = 0$$

$\therefore \sum c_i v_i \in \ker T$ Recall $k_1, \dots, k_s$ is a basis for $\ker T$

$$\therefore \underbrace{\sum_{i=1}^r c_i v_i = \sum_{j=1}^s a_j k_j}_{\text{linear indep among } v_i\text{'s \& } k_j\text{'s}} \left. \begin{array}{l} \text{But } v_i\text{'s and } k_j\text{'s together form} \\ \text{a basis of } V : \\ \text{all } c_i\text{'s and all } a_j\text{'s} = 0 \end{array} \right\}$$

This entire page (a, b, c below) is a very important

More "abstract nonsense" about linear maps Exercise

0 map

(a) Following maps are obviously linear • Any $V \rightarrow W$
 $v \mapsto 0$

• $V \xrightarrow{\text{id}} V$, more generally if W is a subspace of V
 $v \mapsto v$ then the inclusion map $W \rightarrow V$
 $w \mapsto w$

• Composition of linear maps: $V \xrightarrow{f} W \xrightarrow{g} U$ with f, g linear
 $\Rightarrow V \xrightarrow{g \circ f} U$ is linear

(b) We can add linear maps $T_1, T_2: V \rightarrow W$ with same domain & codomain

$$"T_1 + T_2"(v) := T_1(v) + T_2(v) \quad \text{for any } v \in V$$

• We can also scale a linear map $T: V \rightarrow W$
 $"cT"(v) = c(T(v))$

i.e. we add and scale "pointwise".

Check that (i) cT & $T_1 + T_2$ as defined above are linear

(ii) Under these operations, the set of all
linear maps from fixed V to fixed W
is a vector space! denoted $\text{Hom}(V, W)$
or $L(V, W)$

(c) Show that $(A_1 + A_2) \circ B = A_1 \circ B + A_2 \circ B$ } where A_1, A_2, B, C
and $C \circ (A_1 + A_2) = C \circ A_1 + C \circ A_2$ } are linear
 $X \xrightarrow{B} V \xrightarrow{A_1, A_2} W \xrightarrow{C} U$

Claim (We know that any $q \times p$ matrix A defines a linear map $\mathbb{R}^p \rightarrow \mathbb{R}^q$
 $x \mapsto Ax$)

Conversely, any linear map T from $\mathbb{R}^p$ to $\mathbb{R}^q$ is of this form i.e. $T(x) = Ax$ for a unique matrix A .

Pf How to get A from T ? $\nearrow e_1$ $\nearrow e_1$ $\nearrow e_n$
 Take $x = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$ and notice $x = c_1 \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \dots + c_p \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix}$
 $= c_1 e_1 + c_2 e_2 + \dots + c_p e_p$

As T is linear $T(x) = T(\sum c_i e_i)$
 $= \sum c_i T(e_i)$ Let $T(e_i) = u_i = \begin{bmatrix} 1 \\ u_i \\ 1 \end{bmatrix}$
 $= c_1 \begin{bmatrix} 1 \\ u_1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ u_2 \\ 1 \end{bmatrix} + \dots + c_p \begin{bmatrix} 1 \\ u_p \\ 1 \end{bmatrix}$
 Notice that this is $= \begin{bmatrix} 1 & 1 & \dots & 1 \\ u_1 & u_2 & \dots & u_p \\ 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix}$

We have our answer! $A =$ matrix whose columns are $T(e_1), T(e_2), \dots, T(e_p)$ in that order! done
 We will call this A the matrix of T (wrt to standard basis)

Essential

Exercise Observe that following maps are linear for geometric reasons. Find "their matrices"

(i) $\mathbb{R}^2 \rightarrow \mathbb{R}^2$

rotation by θ

(ii) $\mathbb{R}^3 \rightarrow \mathbb{R}^3$

rotation by θ around z axis

(iii) $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ reflection in the line $y=x$

(iv) $\mathbb{R}^3 \rightarrow \mathbb{R}^3$ reflection in XY plane

(v) composition (i) followed by (iii)

Tomorrow: See how linear map $V \xrightarrow{T} W$ for any f.dim'l V, W
can be expressed using a matrix by
fixing a basis for V and a basis for W .